

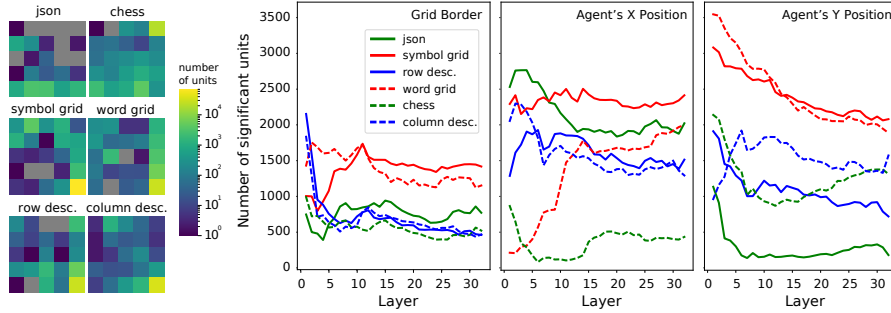


Fig. 5. Supplementary results from probing analysis. Left-most panel shows six heatmaps displaying the number of units that were significantly correlated with the position of the agent being a specific grid cell, for each SIR type. Gray cells denote places where no units were significantly correlated with that position. Note the color scale is logarithmic. The last three panels show the number of parameters per layer significantly correlated with a spatial feature of interest, for each SIR type. The features displayed are whether the agent is located on the grid border, the agent’s x coordinate and the agent’s y coordinate, respectively.

C.4 Ablation Experiments

In our ablation experiments, we silenced specific groups of units—identified as significantly correlated with spatial features—by applying a binary mask during the model’s forward pass. This procedure effectively zeroed out the contributions of the selected parameters while leaving the remainder of the network intact. We then evaluated the model on the 5×5 GWSOT, always using the JSON SIR, to compare its performance with and without ablation and computed the Success Rate for each condition.

D Additional Results from Probing Experiments

In this appendix, we present supplementary analyses from our probing experiments on the LLaMA-3.1-8B model, focusing on the internal encoding of spatial information. The results summarized in Figure 5 provide further insight into how different SIRs influence the model’s activations.

First, when attempting to identify parameters correlated with specific grid locations, we observed that no individual unit was consistently correlated with the same grid cell across all SIR types. However, for single SIRs, we found that some grid positions were associated with thousands of significantly correlated units, while others had few or none (Fig. 5, left-most panel, heatmaps). Although this pattern varied across different SIRs, it was relatively consistent when comparing SIRs within the same class (compare pairs of heatmaps in the same row). This might indicate differences in how the model processes the 2D environment, or it could simply reflect differences in the textual formulation of each SIR rather than a true abstraction of spatial features.

Table 4. Example Prompt and Response Pairs for Alternative Task

Condition	Prompt	Response
Spatial	You are in a library, you see a table to your left and a bookshelf to your right. You need to find the book which should be on the shelf next to the window.	To find the book, I would walk to the bookshelf next to the window, which is to my right.
Spatial	I’m in my office with my desk in front of me. To the right of the desk is a filing cabinet, and behind me is a bookshelf. I think I left my stapler near the filing cabinet; could you help me locate it?	Since the stapler is near the filing cabinet, and the filing cabinet is to the right of the desk, the stapler should be located to the right of the desk.
Non-Spatial	You are a hardworking person who has been told you often overwork yourself, and you wonder if that leads to burnout.	I often put in long hours, prioritizing work over self-care and relaxation. This constant push to be productive might be contributing to burnout. I’ve noticed I’m more irritable and exhausted lately, and my motivation has been wavering. Maybe it’s time to reevaluate my work-life balance.
Non-Spatial	I live in the jungle, have a long tail, and often swing from trees. I enjoy eating bananas. What animal could I be?	You are likely a monkey.

Furthermore, considering that each layer of the LLaMA model comprises 4096 parameters, it is notable that in certain layers and for specific SIRs more than half of these parameters show significant correlations with a spatial feature (last three panels of Fig. 5). It is important to note that this large proportion of the model’s parameters are likely correlated with the form of the input for each specific SIR (i.e. similar prompts should yield similar activations). If this is the case, the effect should be stronger in more superficial layers, whose representations are closer to the input. This is indeed what we see, in contrast to the distribution of parameters significantly correlated to the spatial feature of interest across all SIRs, which usually peaks in intermediate and deep layers. This suggests that these hundreds to thousands of significantly correlated parameters per layer for single SIRs are not truly encoding spatial information but just representing the input text.

E Alternative Spatial/Non-Spatial Reasoning Task

To investigate if units identified as processing spatial information in the GWSOT are also activated in an unrelated spatial context, we designed an additional task

to compare spatial and non-spatial reasoning. We created a dataset of reasoning prompts that fall into two categories:

- **Spatial Reasoning Examples:** These prompts describe everyday scenarios with explicit spatial cues (e.g., directions, relative positions) and require the model to reason about object locations.
- **Non-Spatial Reasoning Examples:** These prompts focus on abstract or personal reflections, descriptions, or general reasoning that do not emphasize spatial layout.

We used the following system prompt across all examples:

You have been tasked with answering reasoning questions.
Please answer precisely and briefly.

For each prompt, the model generated a response and we extracted corresponding hidden state activations for further analysis. Table 4 shows some examples of prompt/response pairs for the spatial and non-spatial conditions.

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